

Modeling Dynamics: Autoregressive and Cross-Lagged Panel Models

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About me

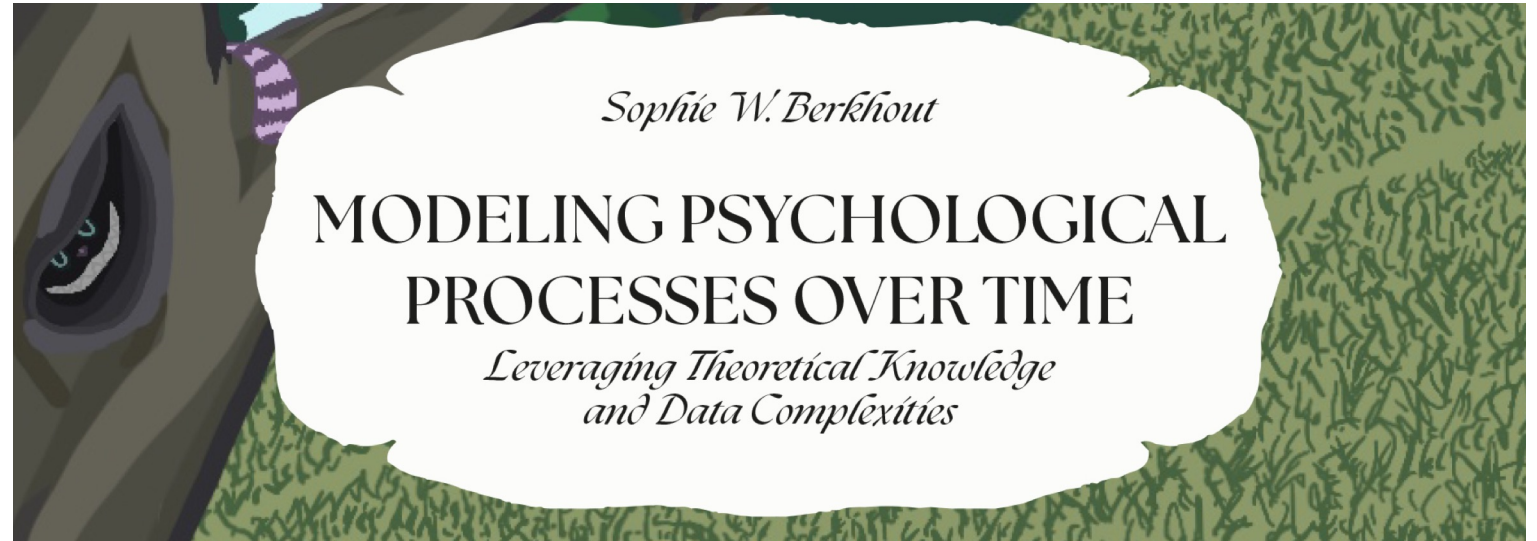
- PhD in Modeling Psychological Processes

Promotors

Ellen Hamaker

Noémi Schuurman

- Associate editor MATILDA
- Analysis Programmer JASP



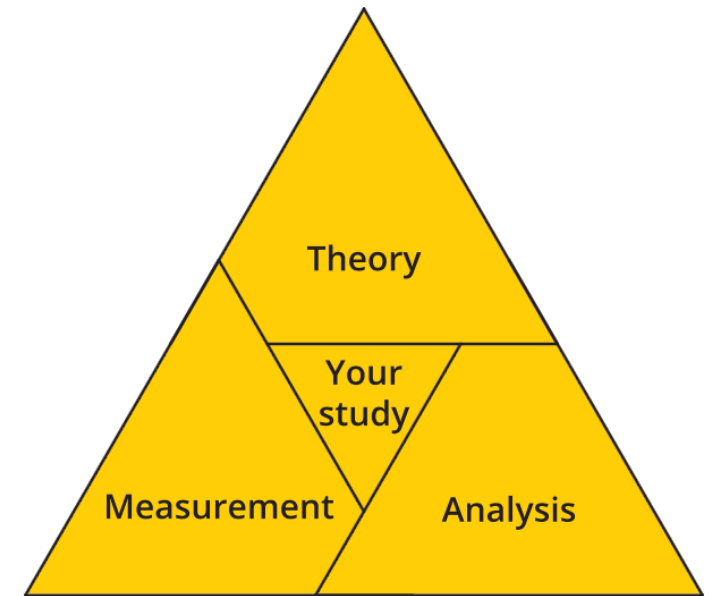
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Measurement, Analysis & Theory for Intensive Longitudinal Data

MATILDA

MATILDA articles highlight key considerations when designing an ILD study.

MATILDA's Model Navigator helps in deciding what models may be most useful in your empirical study.

To help you navigate modeling possibilities

you are asked to think about the temporal pattern that characterizes your process.

Example of an article:

Description, prediction, causation

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This article describes how research questions can be categorized as pertaining to one of three different scientific goals: Descriptive research, predictive research, or causal research. Having a clear understanding of your scientific goal is important as it shapes the subsequent steps in your research process, such as decisions regarding study design, choice of variables, measurement, and statistical analyses (Shmueli, 2010). In this article, we also relate the three different scientific goals to important methodological considerations in ILD research.

Mulder, J. D. (2025). Description, prediction, causation. *MATILDA Preprints*, 2025-05-23. <https://matilda.fss.uu.nl/articles/description-prediction-causation.html>

Overview of this workshop

Day 1: $N = 1$

- Autoregressive model
- Vector autoregressive model

Day 2: $N > 1$

- Multilevel autoregressive model
- Cross-lagged panel model (CLPM)
- Random-intercept CLPM

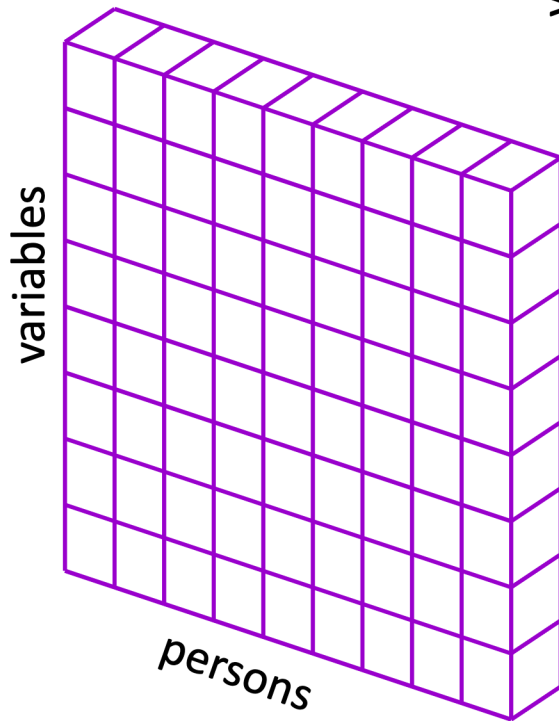


Materials

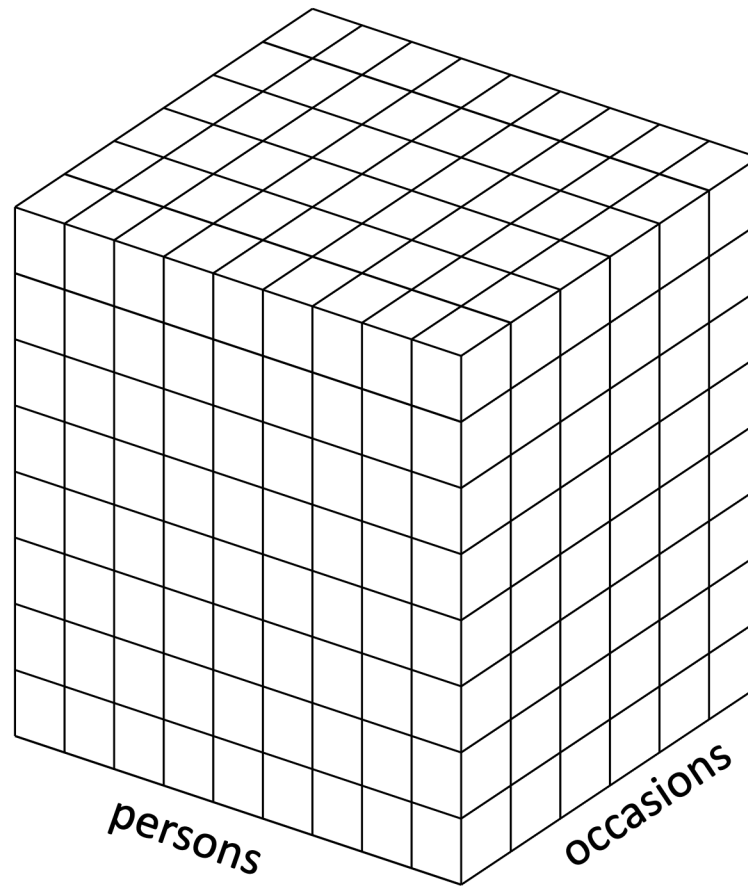
Slides and exercises are available at:

<https://github.com/sophieberkhout/modeling-dynamics-workshop>

**Inter-individual or
cross-sectional data**



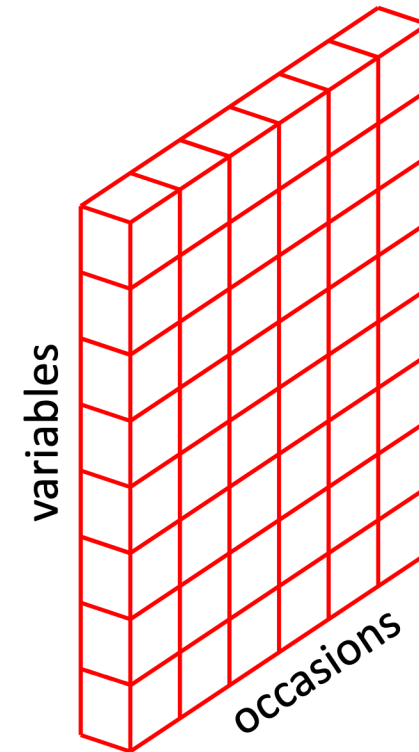
variables



persons

occasions

**Intra-individual or
time series data**



variables

occasions

Cattell's data box

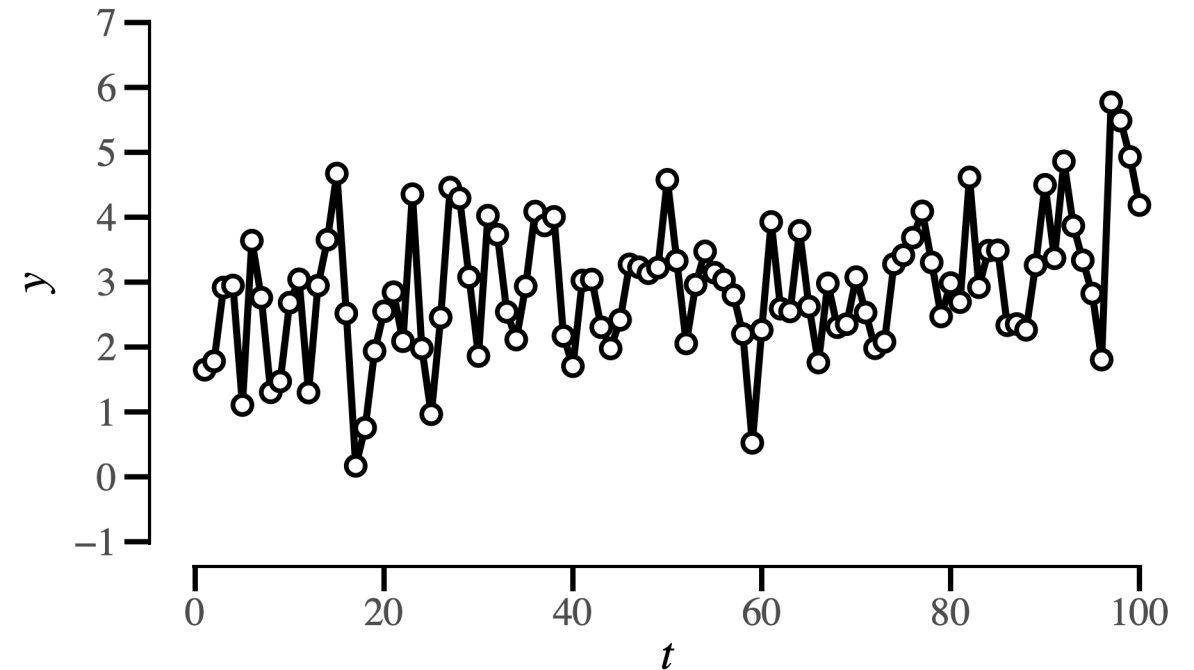
From Hamaker, E. L. (2025). Common data types: Cross-sectional, time series, panel, and intensive longitudinal data. *MATILDA Preprints*, 2025-05-23.

<https://matilda.fss.uu.nl/articles/common-data-types.html>

Intensive longitudinal data (ILD)

- Daily diary
- Experience Sampling Method (ESM)
- Ecological momentary assessment
- Ambulatory assessment

y_t	← Data is in long format
y_1	
y_2	
y_3	
y_4	
y_5	
y_T	



Why model dynamics?

How does a process unfold over time?

Autoregressive and cross-lagged effects are popular dynamics in the psychological literature.

E.g., emotional inertia

Kuppens, P., Allen, N. B., & Sheeber, L. B. (2010). Emotional Inertia and Psychological Maladjustment. *Psychological Science*, 21(7), 984–991.

<https://doi.org/10.1177/0956797610372634>

The first-order autoregressive model

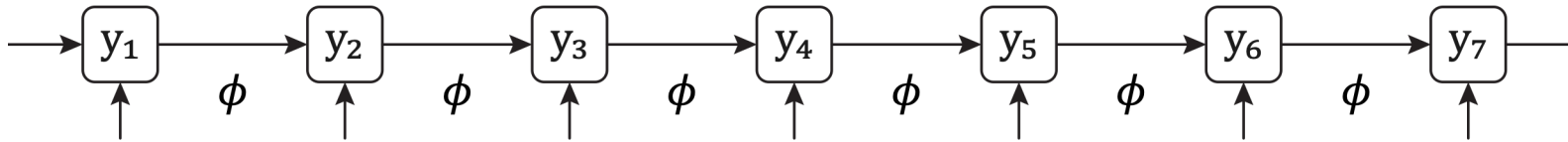
First-order autoregressive model (AR[1])

How does a process carryover from one occasion to the next?

$$\cdots \rightarrow y_{t-1} \rightarrow y_t \rightarrow y_{t+1} \rightarrow \cdots$$

First-order / (1) means steps of one occasion or a lag of 1

First-order autoregressive model (AR[1])

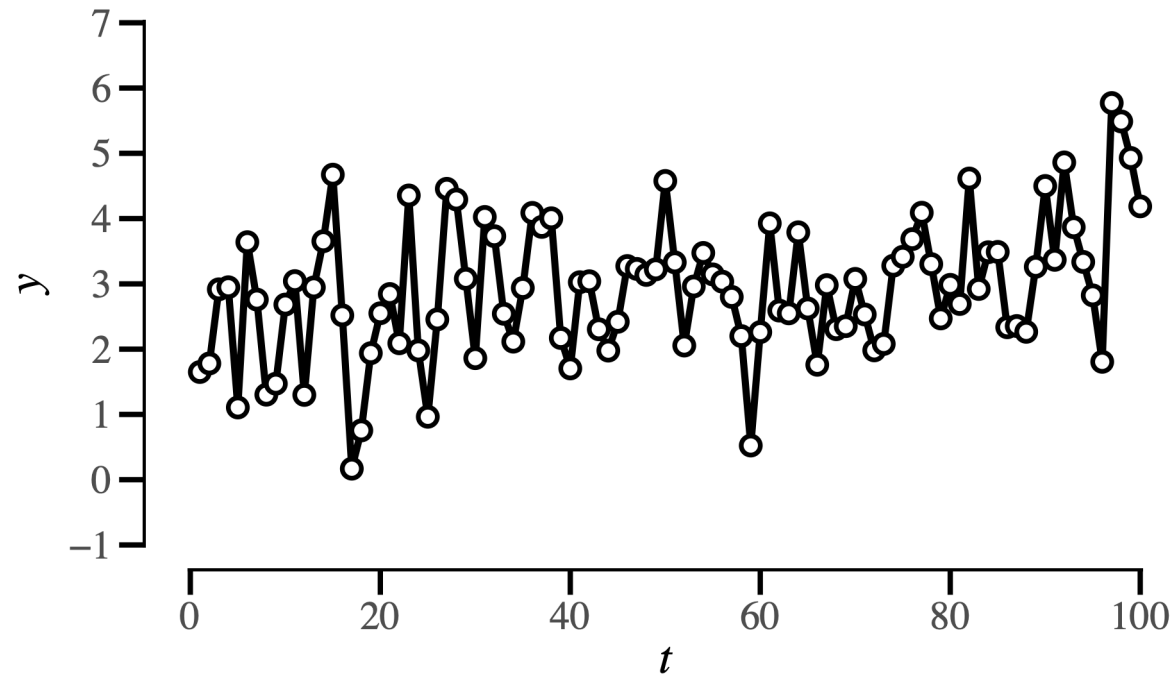


$$y_t = c + \phi \cdot y_{t-1} + \varepsilon_t$$
$$\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

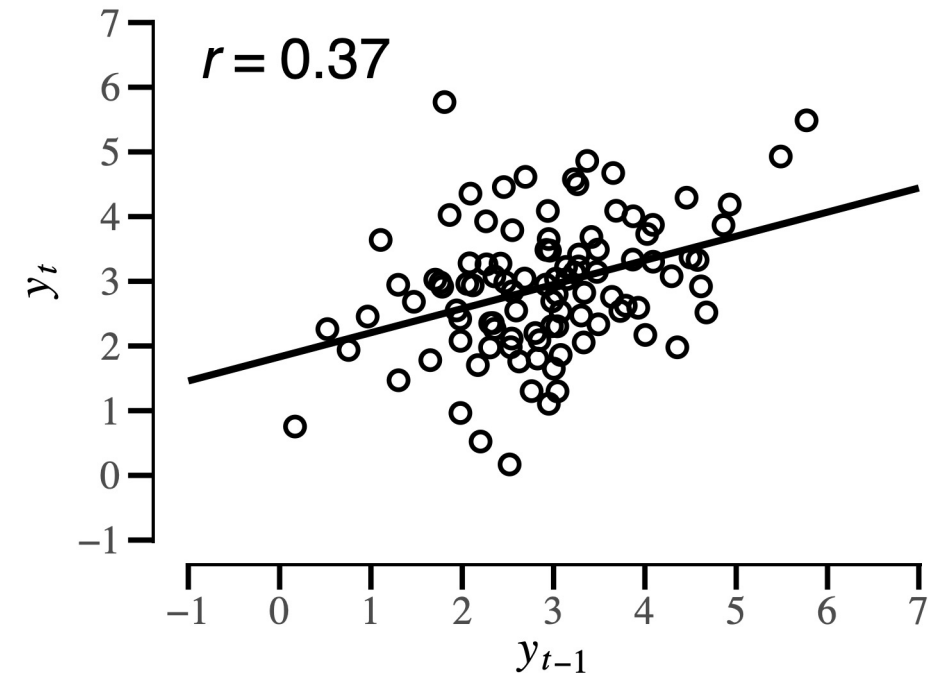
Look familiar?

AR(1) visualized

Time series plot



Autocorrelation scatter plot



Data format

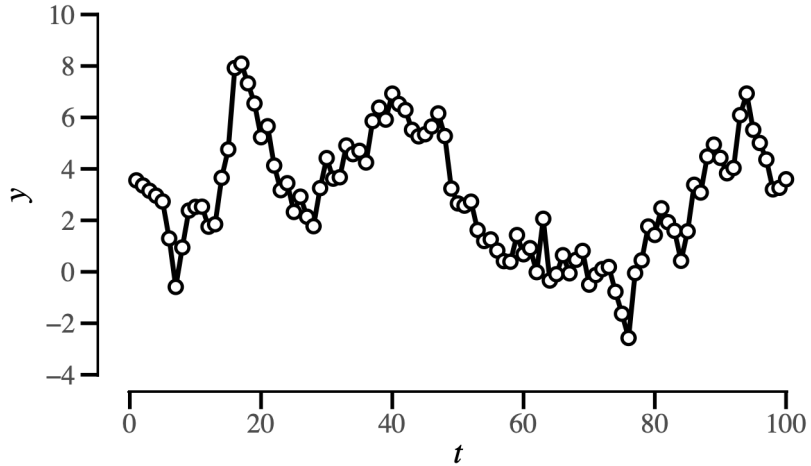
You never have an observation at $t = 0$.

→ Analysis sample size will be $T - 1$.

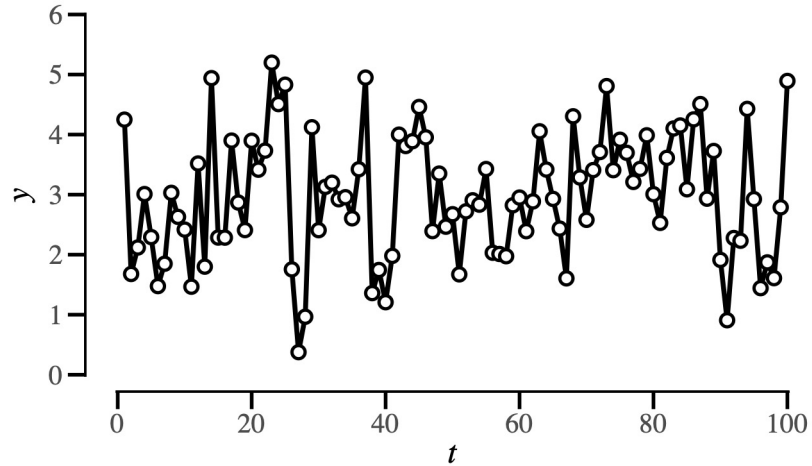
t	y_t	y_{t-1}
1	y_1	NA
2	y_2	y_1
3	y_3	y_2
4	y_4	y_3
5	y_5	y_4
T	y_T	y_{T-1}

Autoregression strength and direction

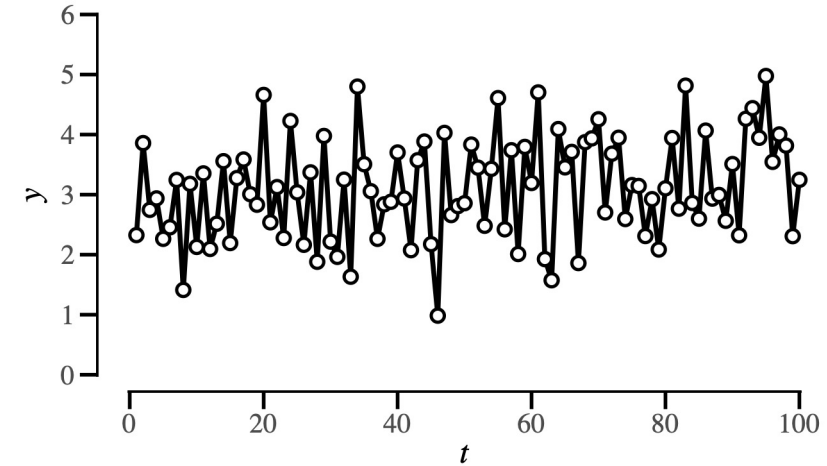
$$\phi = 0.9$$



$$\phi = 0.3$$



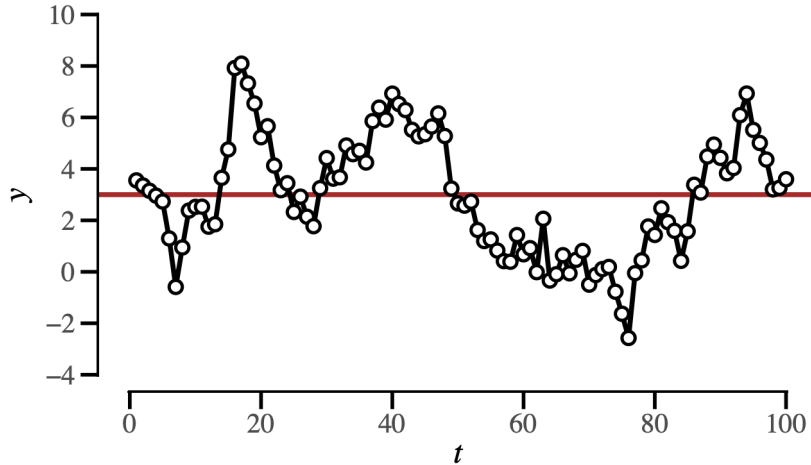
$$\phi = -0.3$$



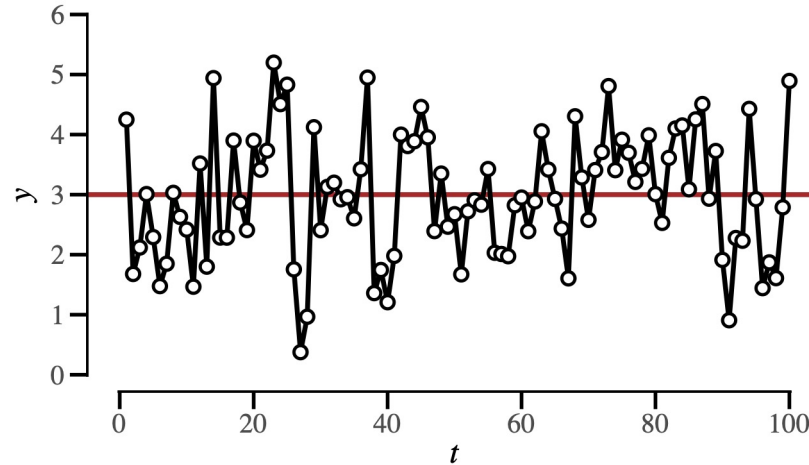
Autoregression strength and direction

The mean matters!

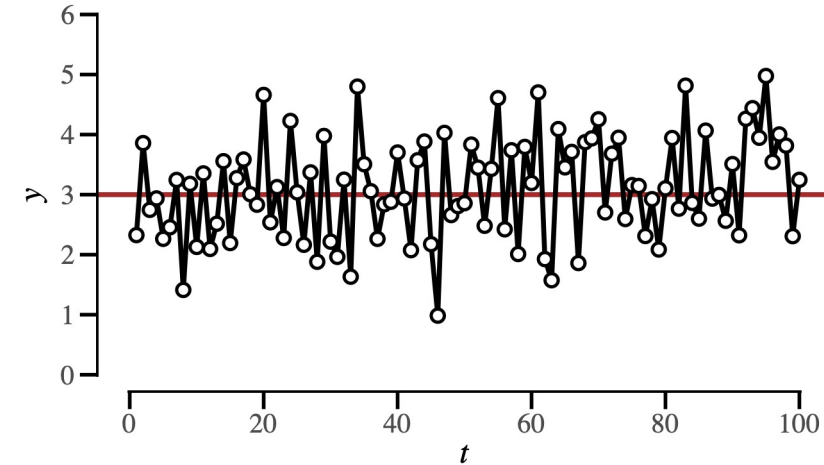
$$\phi = 0.9$$



$$\phi = 0.3$$



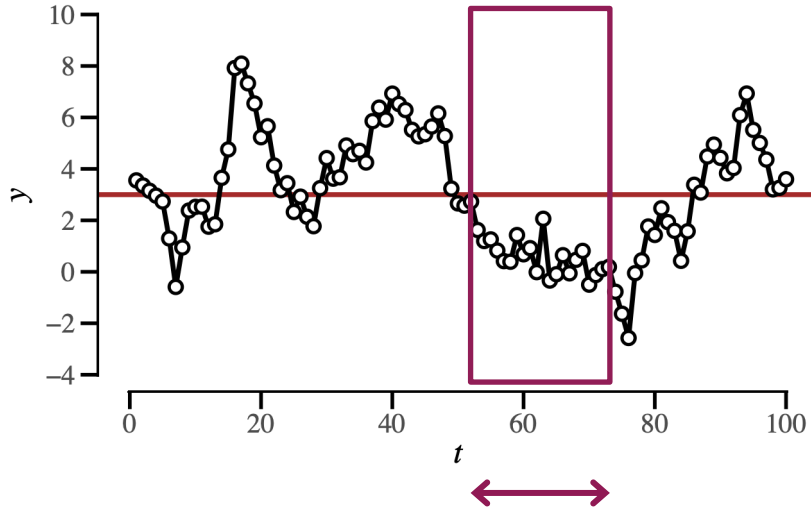
$$\phi = -0.3$$



Autoregression strength and direction

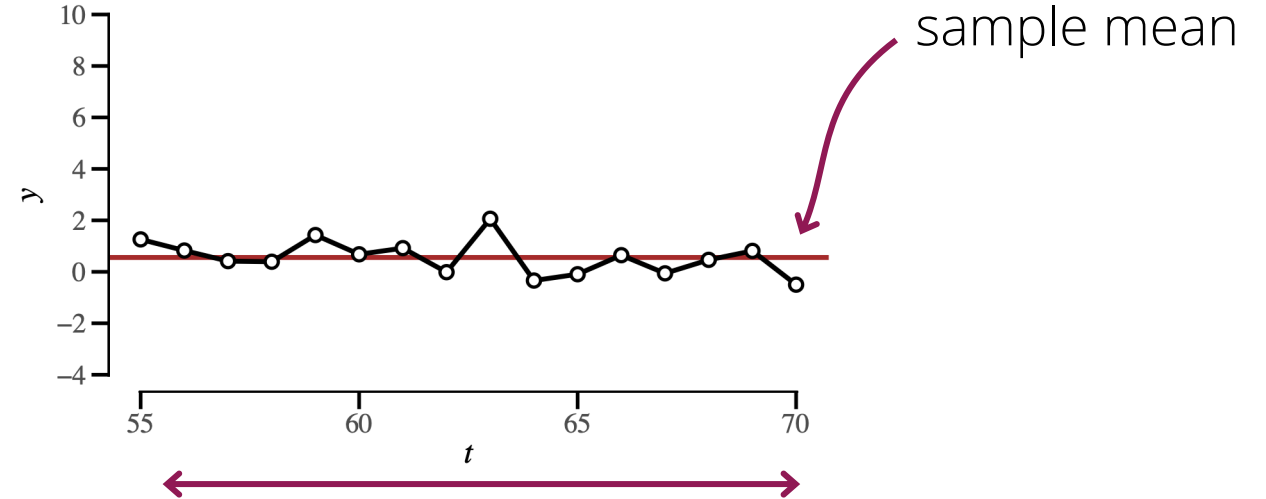
The mean matters!

$$\phi = 0.9$$



The time span of the study matters!

$$\phi = -0.3$$



Relation between mean, intercept, and autoregression

The mean of the autoregressive model is a function of the intercept and the autoregressive parameter.

$$\mu = \frac{c}{1 - \phi}$$

$$c = (1 - \phi) \mu$$

$$\phi = 1 - \frac{c}{\mu}$$

Exercise: Introduction to AR(1)

Playing around with data simulation

File: intro_ar.Rmd

Alternative formulation AR(1) model

The mean is often more informative.

→ Center the predictor.

$$y_t = c + \phi \cdot y_{t-1} + \varepsilon_t$$

$$= \mu + \phi \cdot (y_{t-1} - \mu) + \varepsilon_t$$

(preferably the latent mean over sample mean,
in case of small sample sizes, missing data, or
multilevel modeling)*

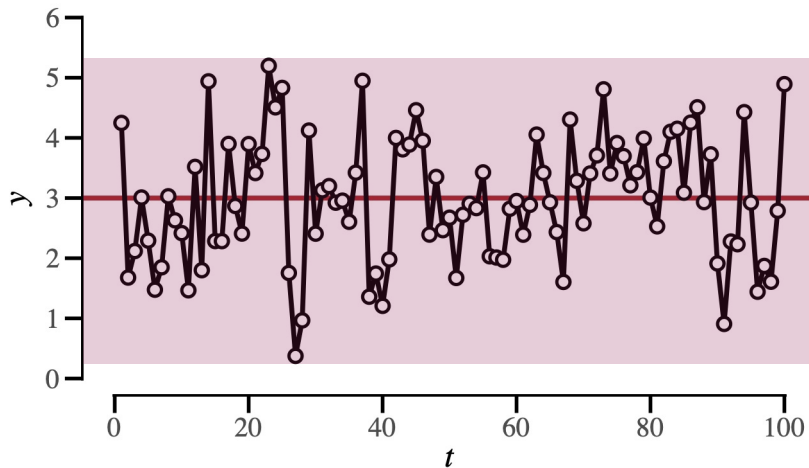
*Lüdtke, O., Marsh, H. W., Robitzsch, A., Trautwein, U., Asparouhov, T., & Muthén, B. (2008). The multilevel latent covariate model: A new, more reliable approach to group-level effects in contextual studies. *Psychological Methods*, 13(3), 203–229. <https://doi.org/10.1037/a0012869>

Two important model assumptions

1. Stationarity

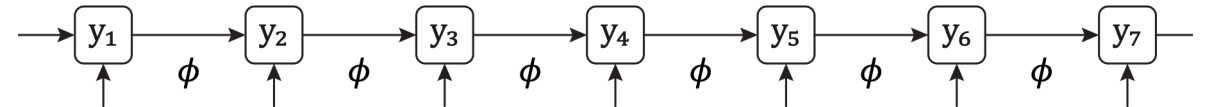
The statistical properties are stable over time

→ ϕ has to lie between -1 and 1



2. Equally spaced observations

The time intervals between observations are equal

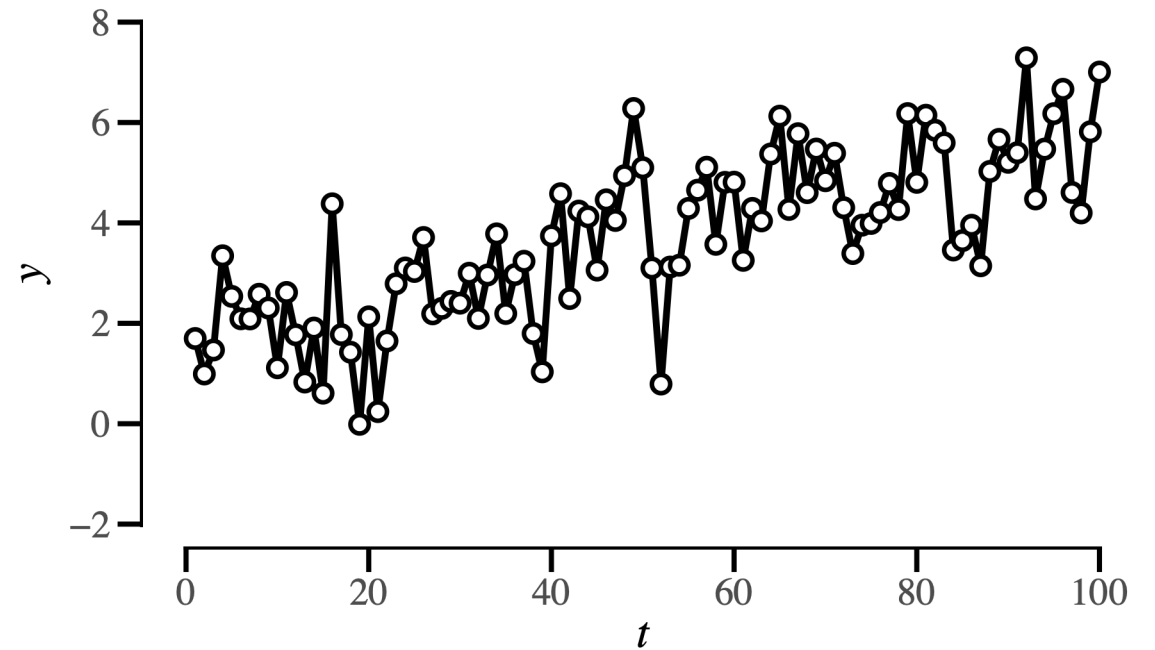


1. Stationarity

Different forms of non-stationarity

- Trend
- Regime-switching
- Time-varying parameters
- Day effects

→ Use a model that can account for it.

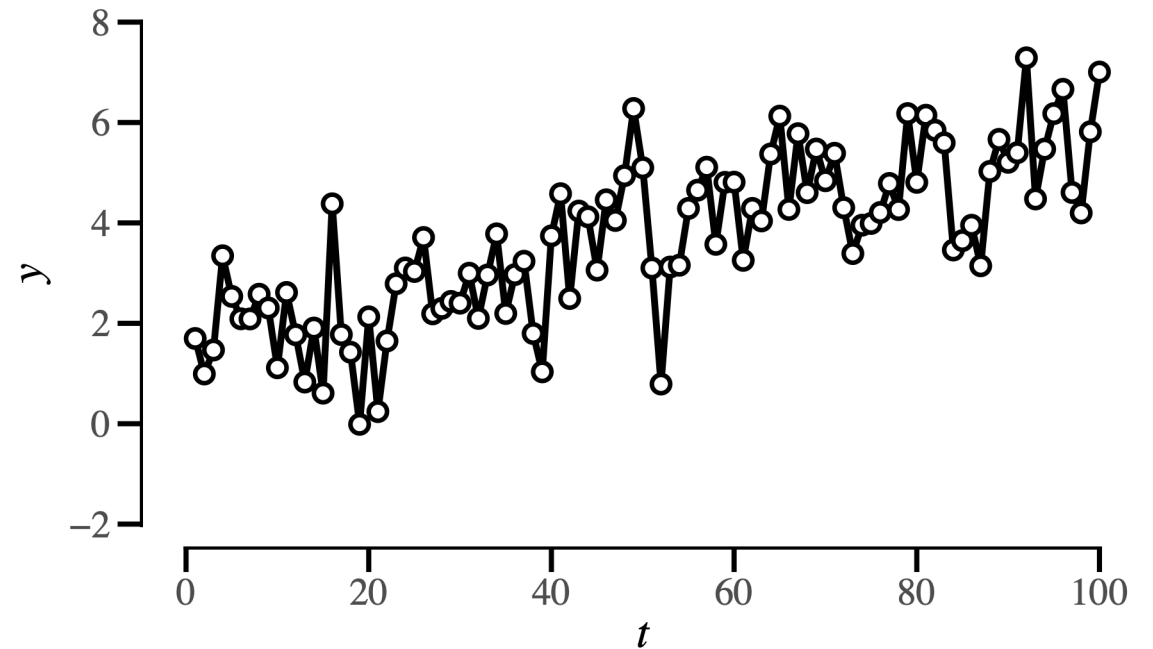


1. Stationarity

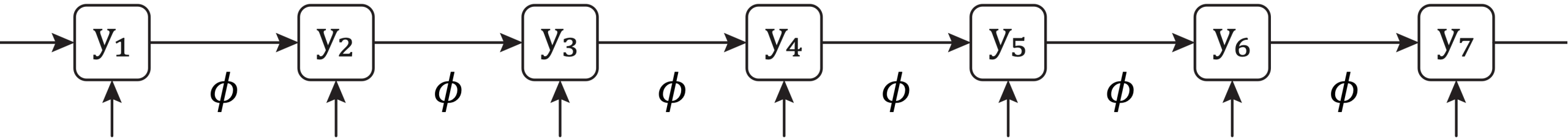
Different forms of non-stationarity

- Trend
→ differencing vs. deterministic trend
- Regime-switching
→ e.g., Markov-switching, threshold, change point
- Time-varying parameters
→ e.g., regression splines, kernel-smoothing
- Day effects
→ e.g., day-level, day predictor

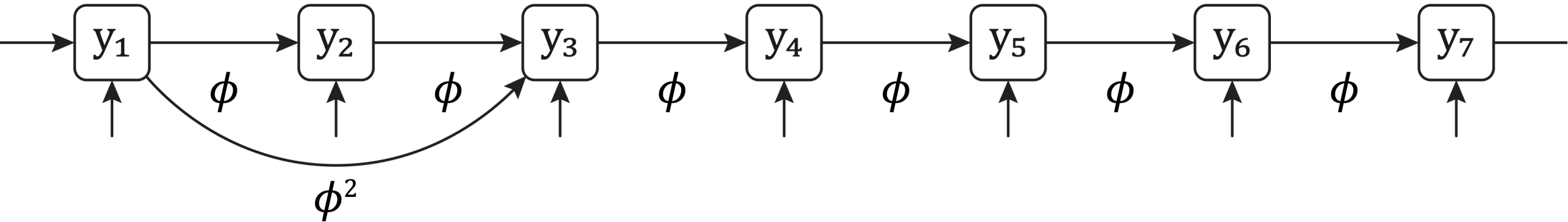
→ Use a model that can account for it.



2. Equally spaced observations



2. Equally spaced observations

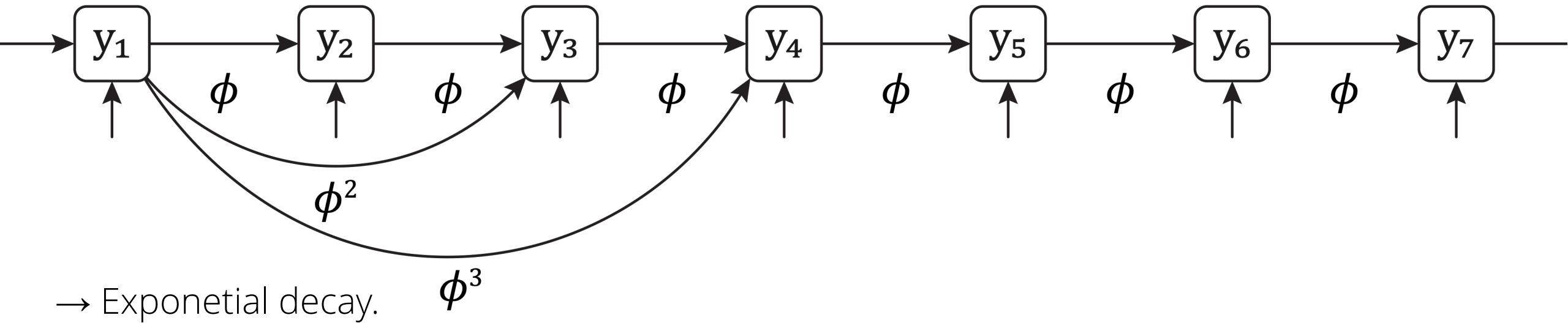


→ Exponential decay.

e.g.,

$$\begin{aligned}\phi &= 0.4 \\ \phi^2 &= 0.16\end{aligned}$$

2. Equally spaced observations



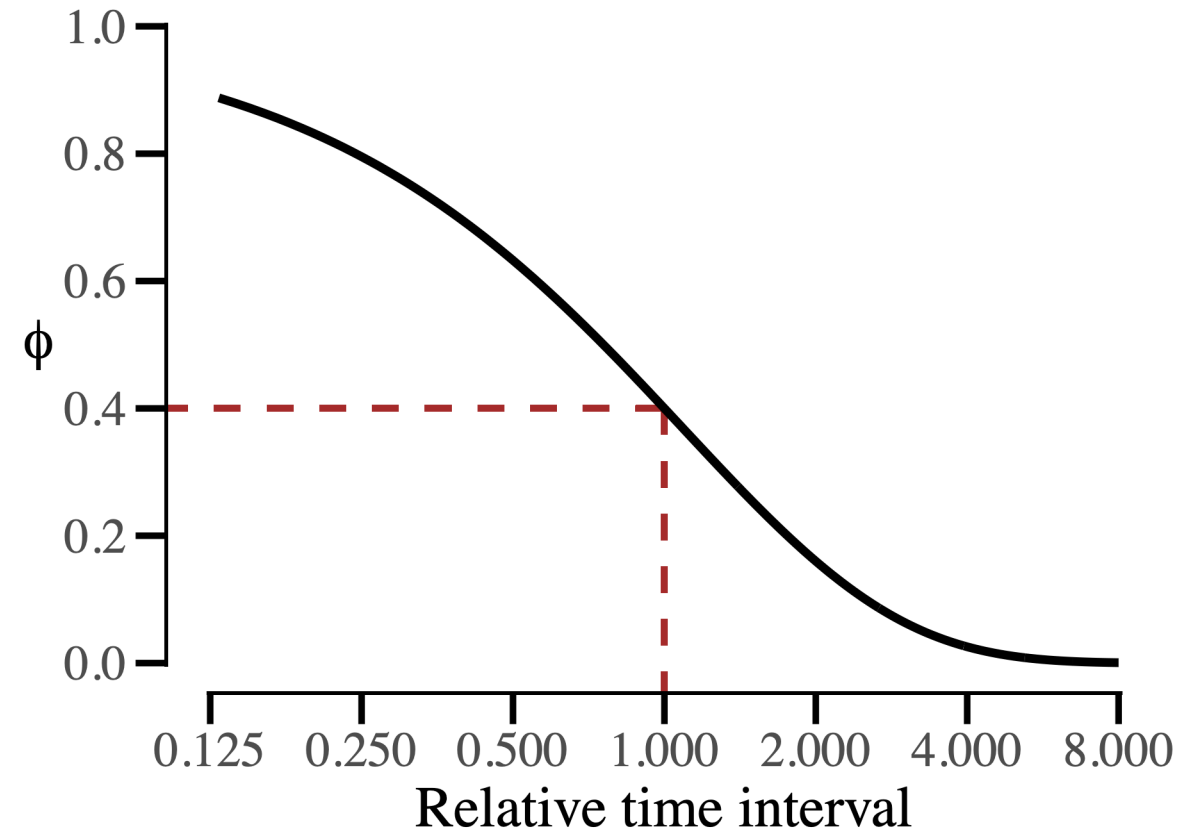
e.g.,

$$\begin{aligned}\phi &= 0.4 \\ \phi^2 &= 0.16 \\ \phi^3 &= 0.064\end{aligned}$$

2. Equally spaced observations

Exponential decay

→ The strength of the autoregressive effect depends on the time interval.



2. Equally spaced observations

→ Be careful with missing data!

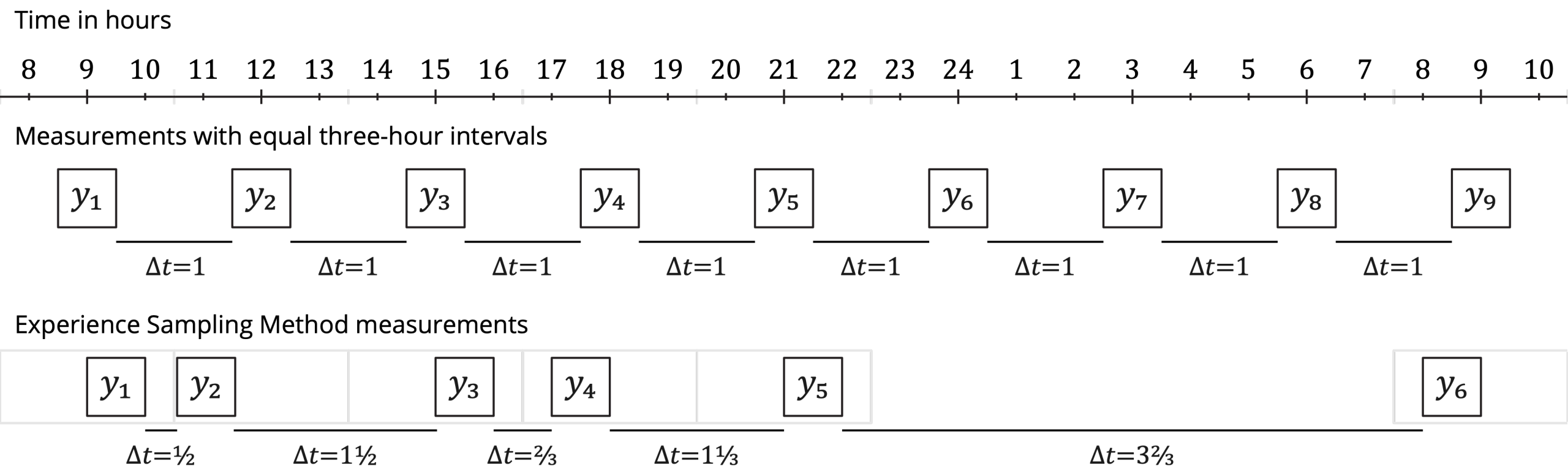
e.g., if y_4 is missing

y_t	y_{t-1}
y_1	NA
y_2	y_1
y_3	y_2
y_5	y_3
y_6	y_5
y_T	y_{T-1}

y_t	y_{t-1}
y_1	NA
y_2	y_1
y_3	y_2
y_4	NA
y_5	y_4
y_T	y_{T-1}

2. Equally spaced observations

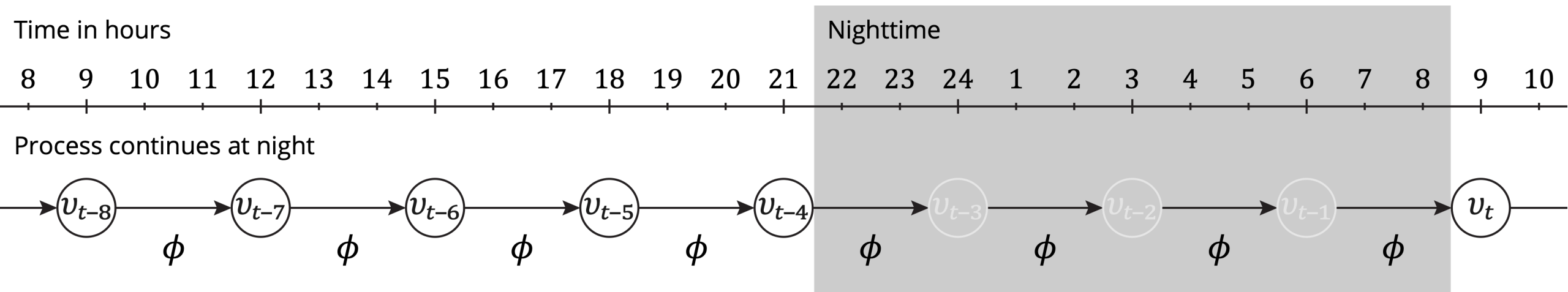
ESM data almost always violates this assumption.



2. Equally spaced observations

→ Treat it as a missing data problem.

e.g., missing night data



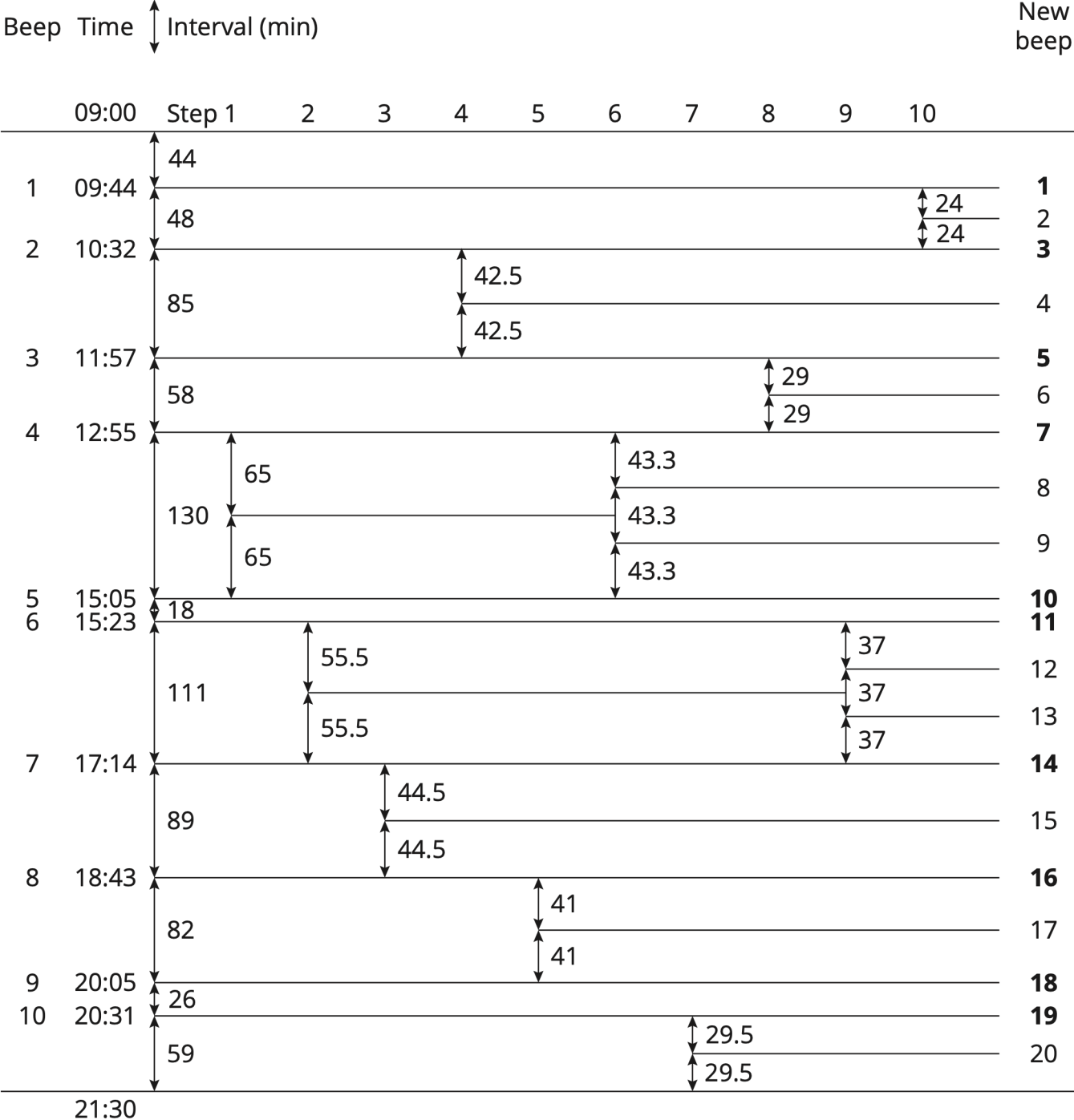
Berkhout, S. W., Schuurman, N. K., & Hamaker, E. L. (2025). Let sleeping dogs lie?
How to deal with the night gap problem in experience sampling method data.
Psychological Methods. <https://doi.org/10.1037/met0000762>

2. Equally spaced observations

→ Treat it as a missing data problem.

e.g., ESM data

Similar to TINTERVAL in Mplus.



2. Equally spaced observations

→ Treat it as a missing data problem.

Requires proper missing data handling in the estimation!

e.g., Kalman filter or Bayesian

<https://matilda.fss.uu.nl/articles/kalman-filter.html>

Gelman, Andrew, J. B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. 2013. Bayesian Data Analysis. Third Edition. London: Chapman & Hall / CRC Press.
Free PDF available: <https://sites.stat.columbia.edu/gelman/book/>

Extensions

- Higher-order AR
<https://matilda.fss.uu.nl/articles/autoregressive-model.html#second-order-autoregressive-model-ar2-model>
<https://matilda.fss.uu.nl/articles/autocorrelation.html>
- Moving-average models
<https://matilda.fss.uu.nl/articles/arma-model.html>
<https://matilda.fss.uu.nl/articles/arima-model.html>
<https://matilda.fss.uu.nl/articles/armax-model.html>
- And many others...
<https://matilda.fss.uu.nl/analysis/model-overview.html>

AR(1) summary

- Autoregressive effect estimates temporal dependency
- Positive AR is often interpreted as inertia (i.e., resistance to change)
- Assumes stationarity and equally spaced observations

Exercise: Fit an AR(1) to empirical data

Working with an empirical data set

File: ar1.Rmd

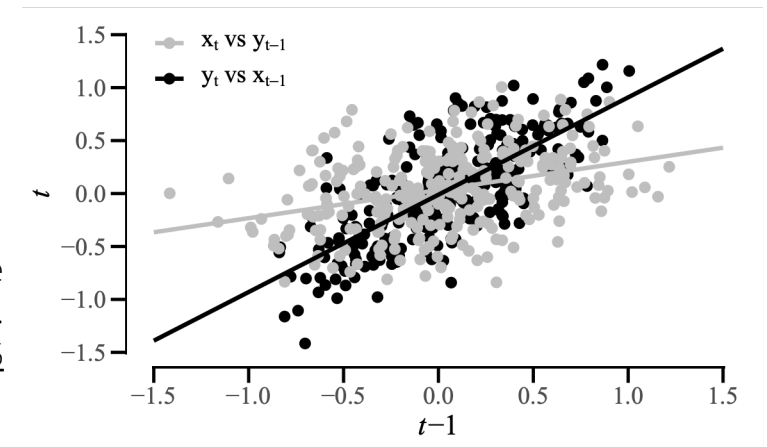
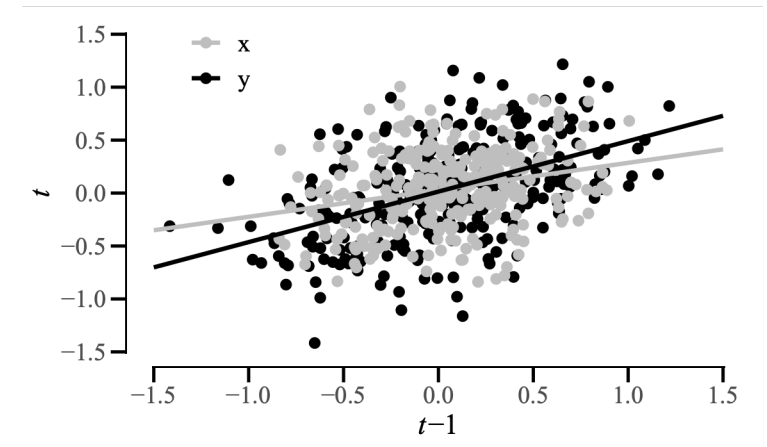
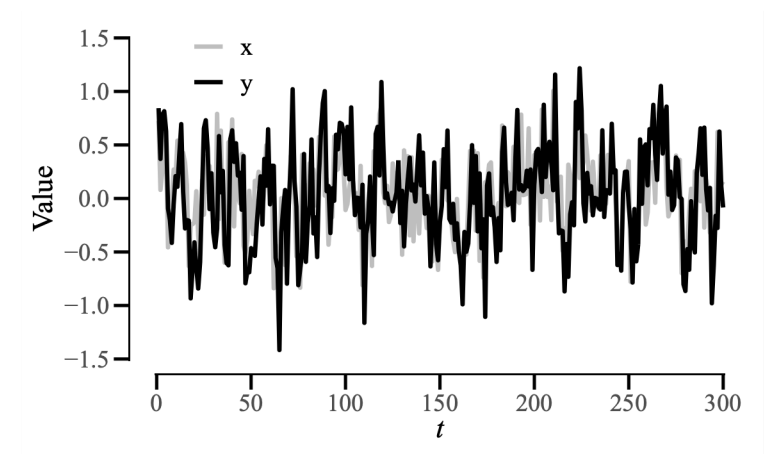
The first-order vector autoregressive model

First-order vector autoregressive model (VAR[1])

Process with two or more variables.

Interest is in the reciprocal relations between them.

E.g., dyadic interactions.



Berkhout, S. W., Schuurman, N. K., & Hamaker, E. L. (2025). A tool to simulate and visualize dyadic interaction dynamics. *Psychological Methods*, 30(3), 599–621.
<https://doi.org/10.1037/met0000575>

First-order vector autoregressive model (VAR[1])

$$y_t = c_y + \phi_y \cdot y_{t-1} + \beta_y \cdot x_{t-1} + \varepsilon_{y,t}$$

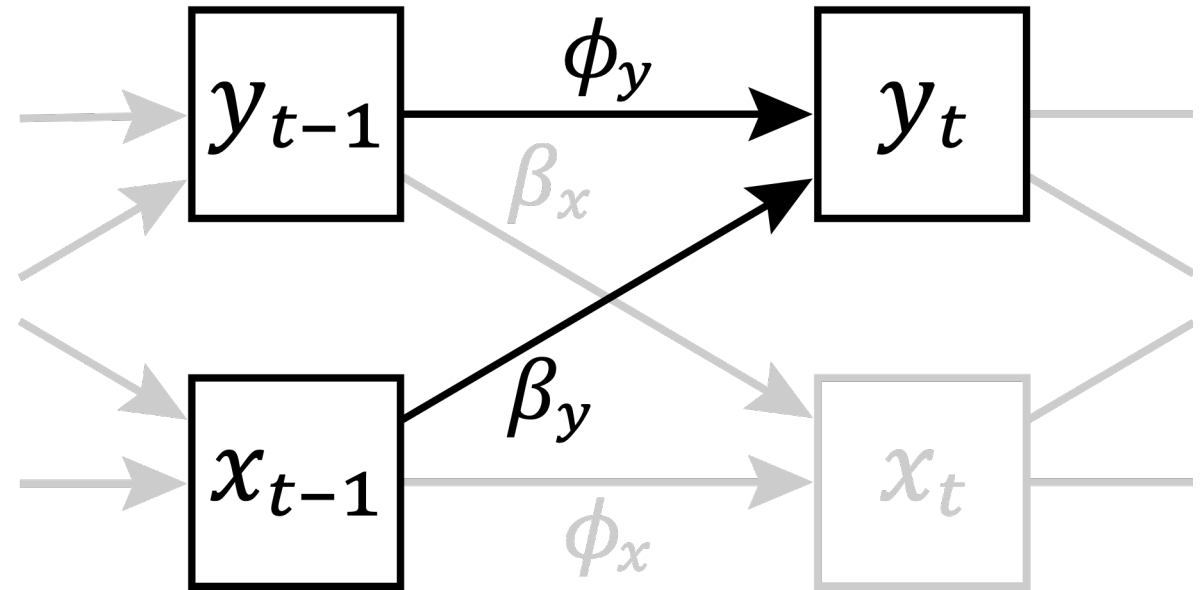
$$x_t = c_x + \phi_x \cdot x_{t-1} + \beta_x \cdot y_{t-1} + \varepsilon_{x,t}$$

$$\varepsilon \sim \mathcal{N}(0, \Sigma)$$

where β_y and β_x are the cross-lagged effects.

Interpretation:

How much one variable spills over to another variable over time.

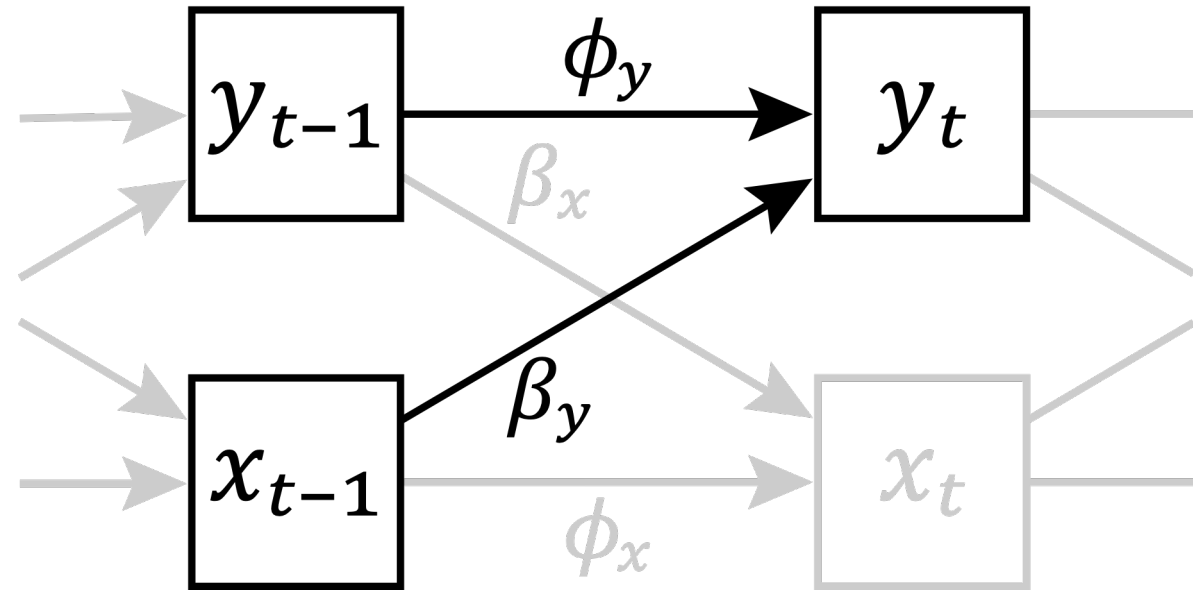


First-order vector autoregressive model (VAR[1])

AKA temporal network analysis.

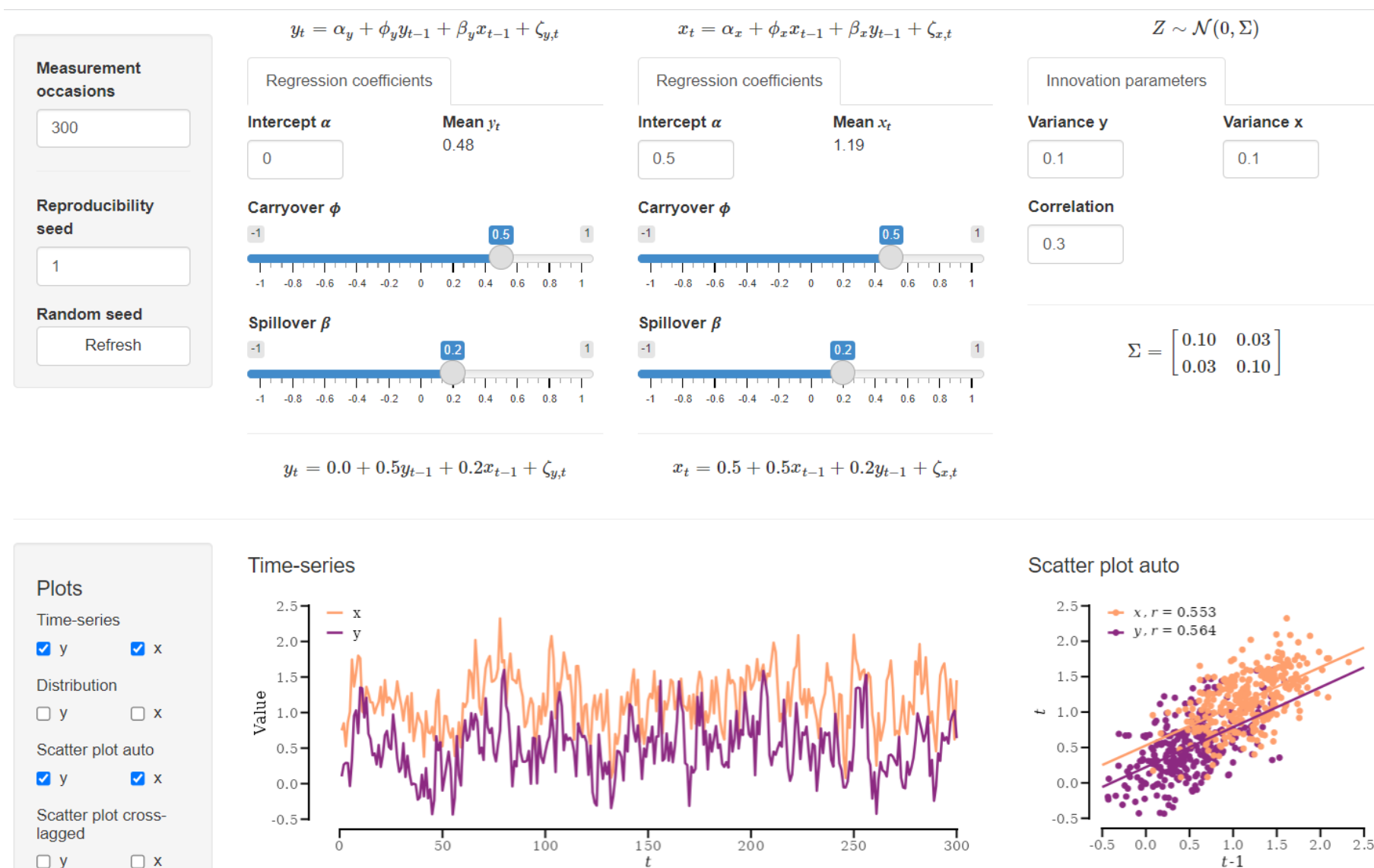
Same assumptions of stationarity and equally spaced intervals.

But also: All variables should be measured at the same time.



Dyadic Interaction Dynamics Shiny App

<https://utrecht-university.shinyapps.io/dyadic-interaction-dynamics/>



VAR(1) estimation

The **vars** package estimates two univariate regression models.

The covariance matrix of the innovations is computed afterwards, and it cannot handle missing values.

For multivariate estimation, you can use the **psychonetrics** package.

Exercise: Fit a VAR(1) model

Same empirical data set, different model

File: var1.Rmd

More resources

- ESM item repository:
<https://esmitemrepositoryinfo.com>
- Open ESM data:
<https://openesmdata.org>
- The Open Handbook of Experience Sampling
Methodology:
<https://www.kuleuven.be/samenwerking/real/real-book>



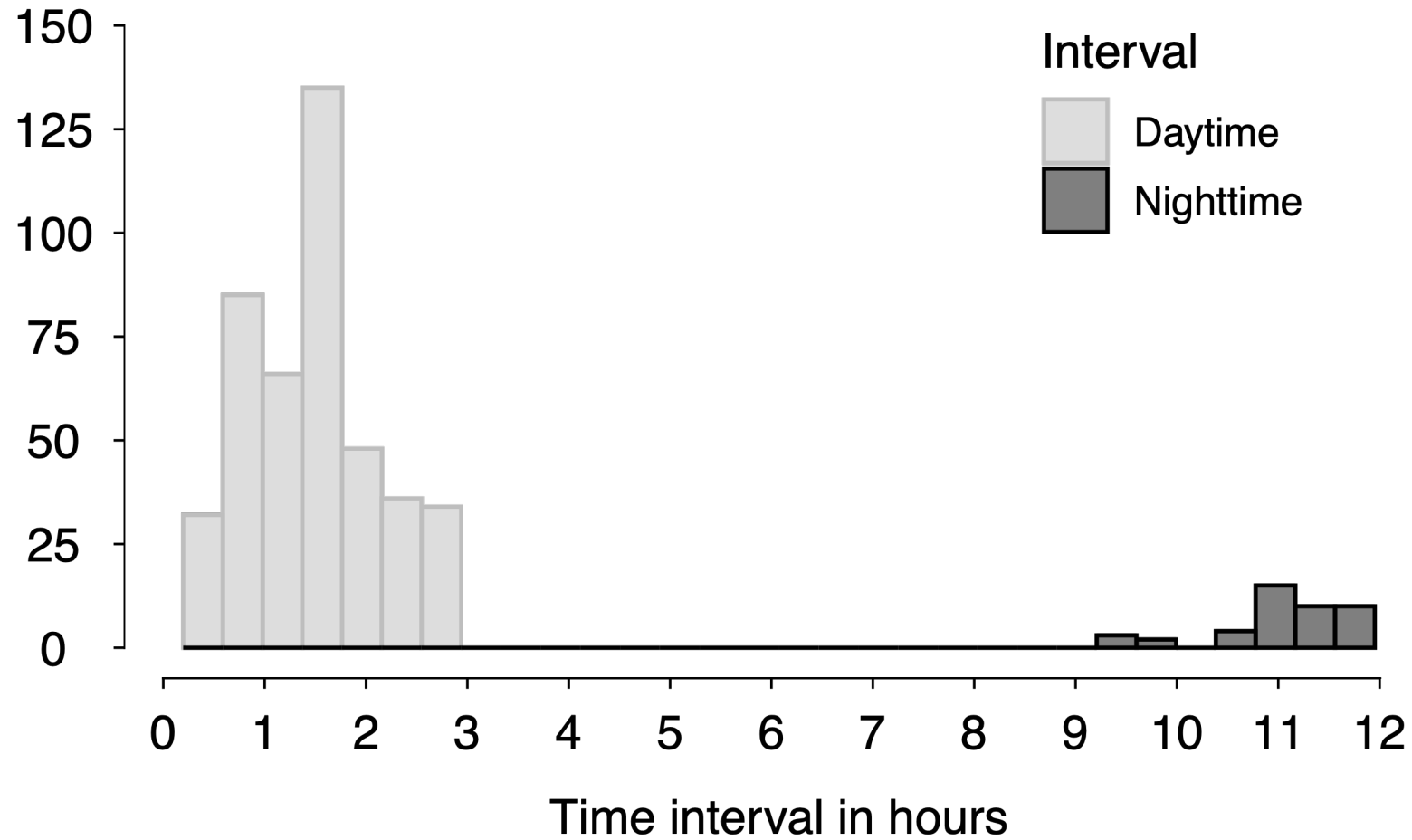
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University**

Sharing science,
shaping tomorrow

Bonus: The night gap problem in ESM data

Berkhout, S. W., Schuurman, N. K., & Hamaker, E. L. (2025). Let sleeping dogs lie? How to deal with the night gap problem in experience sampling method data. *Psychological Methods*. <https://doi.org/10.1037/met0000762>

Long gaps in the data collection at night

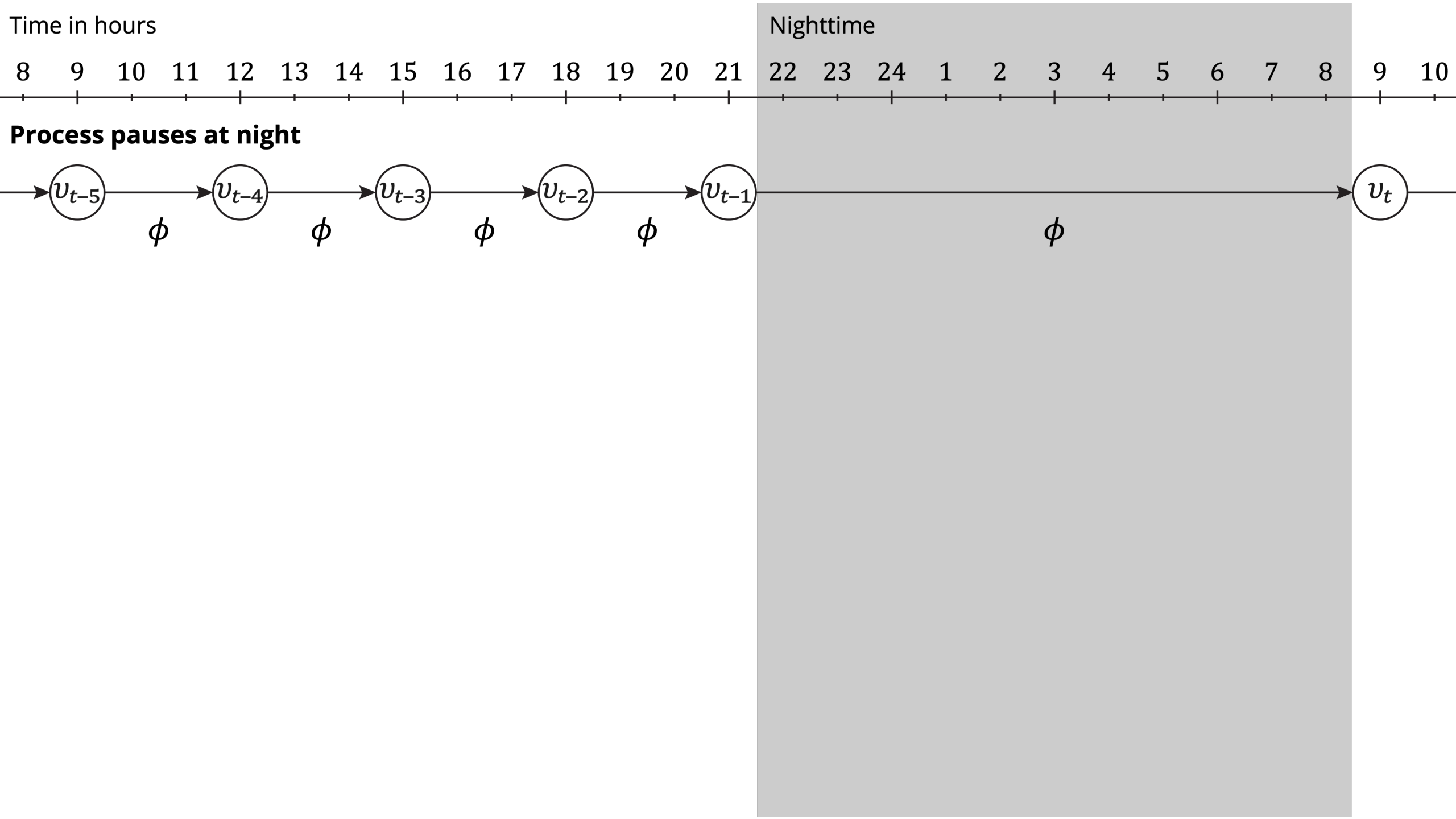


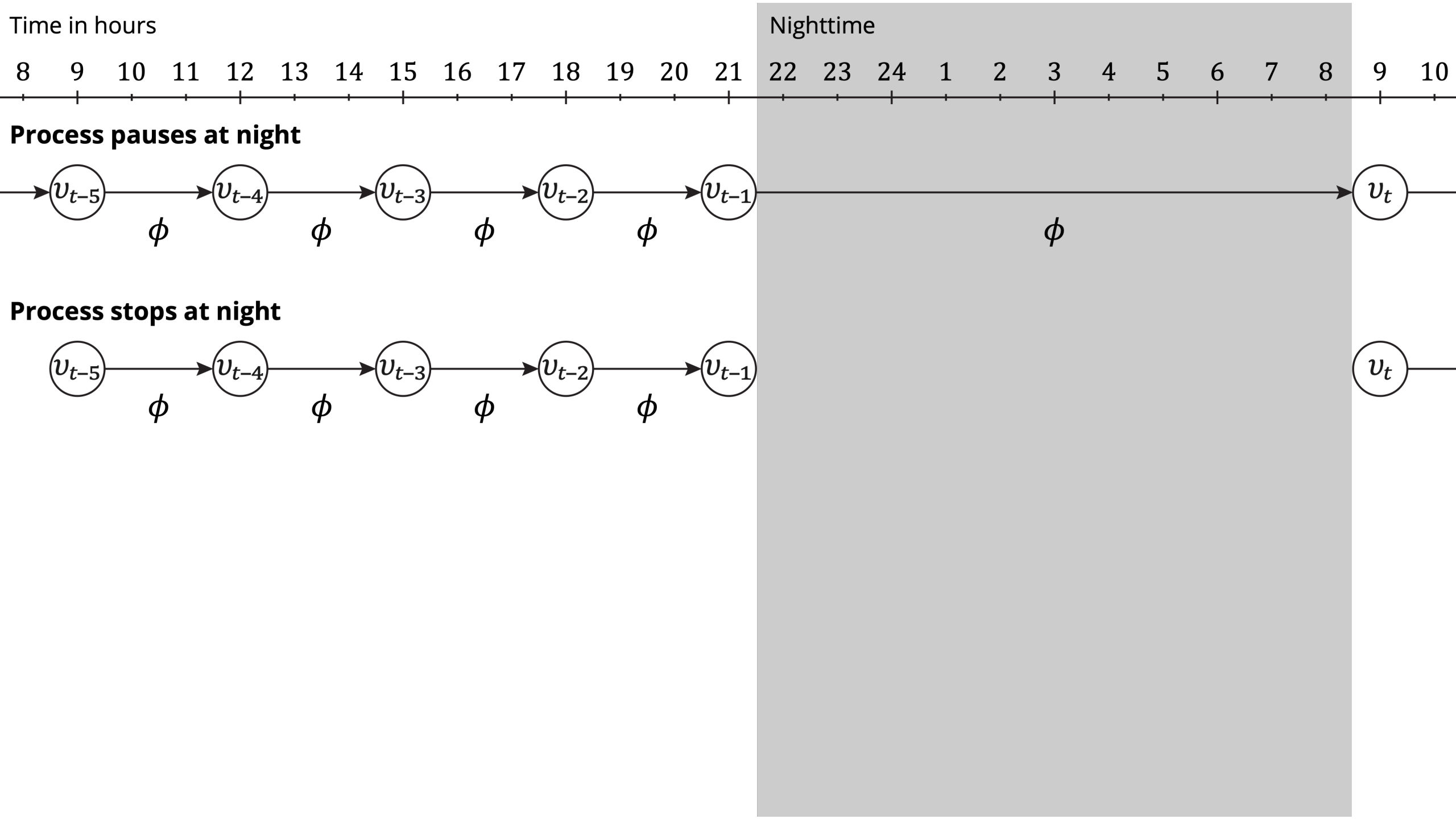
What happens at night?

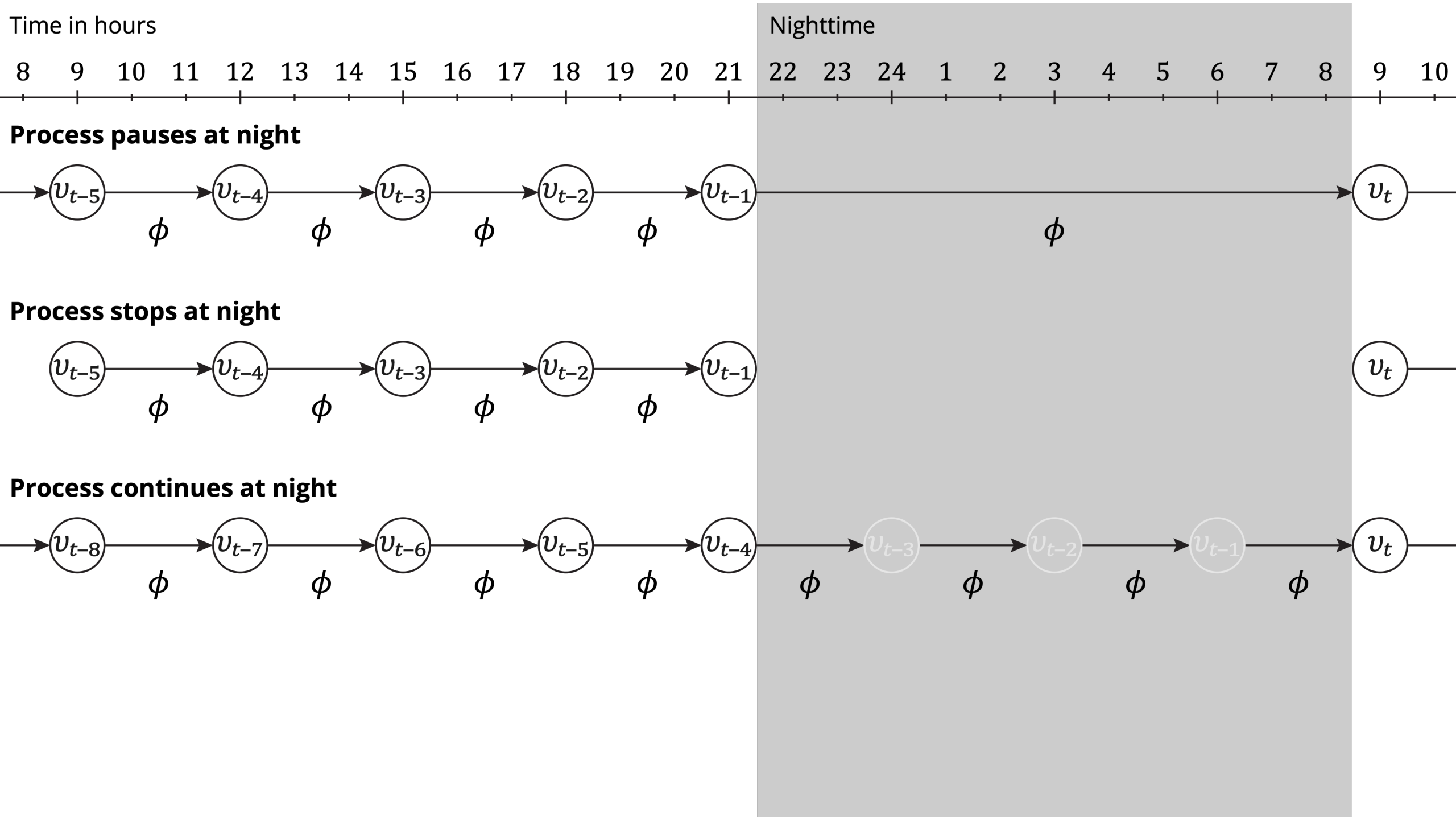
What are the theoretical assumptions for the process of interest?

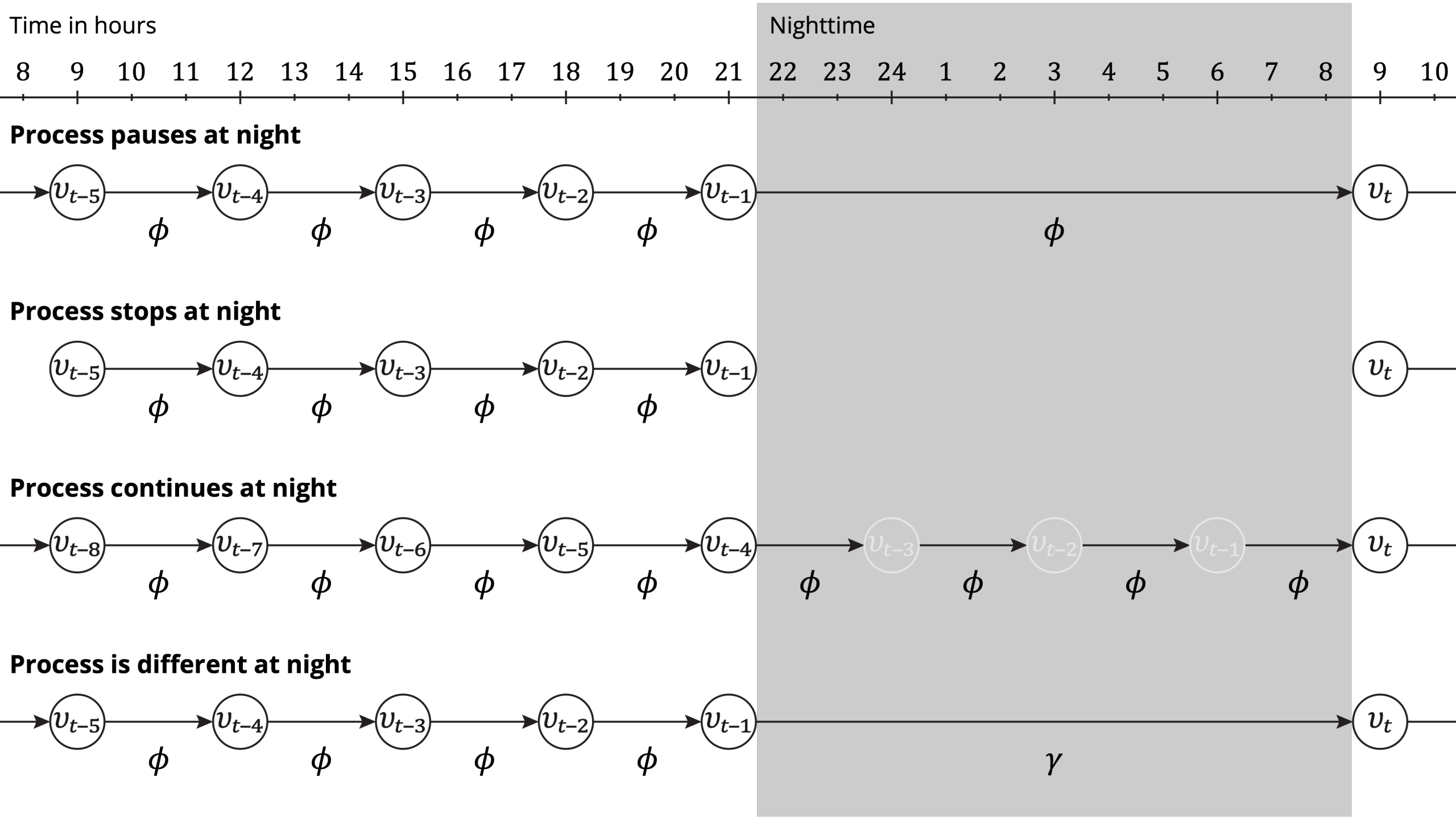
How can we test whether these assumptions are supported by data?

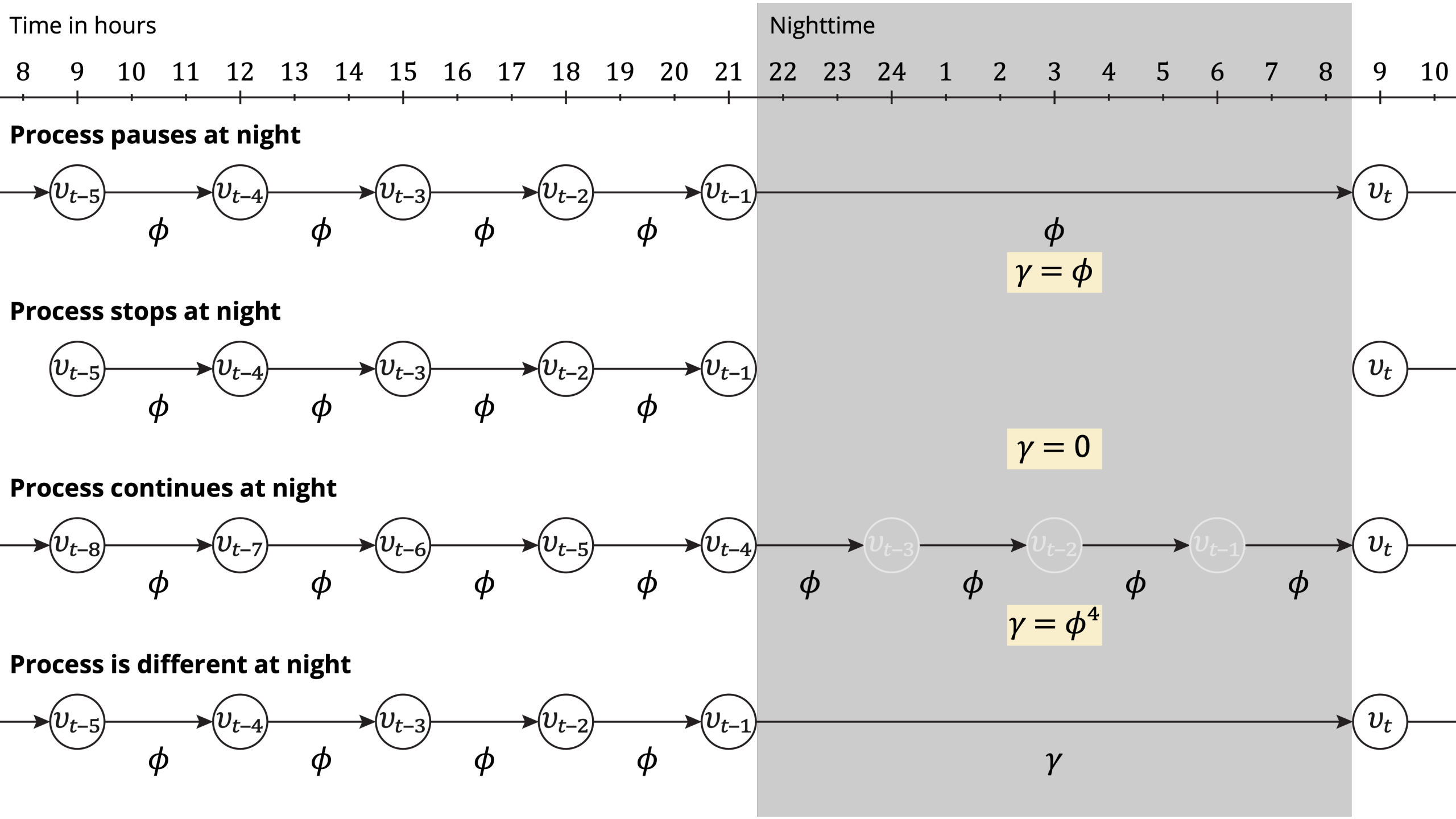
→ Alternative method:
a general unconstrained version of existing methods











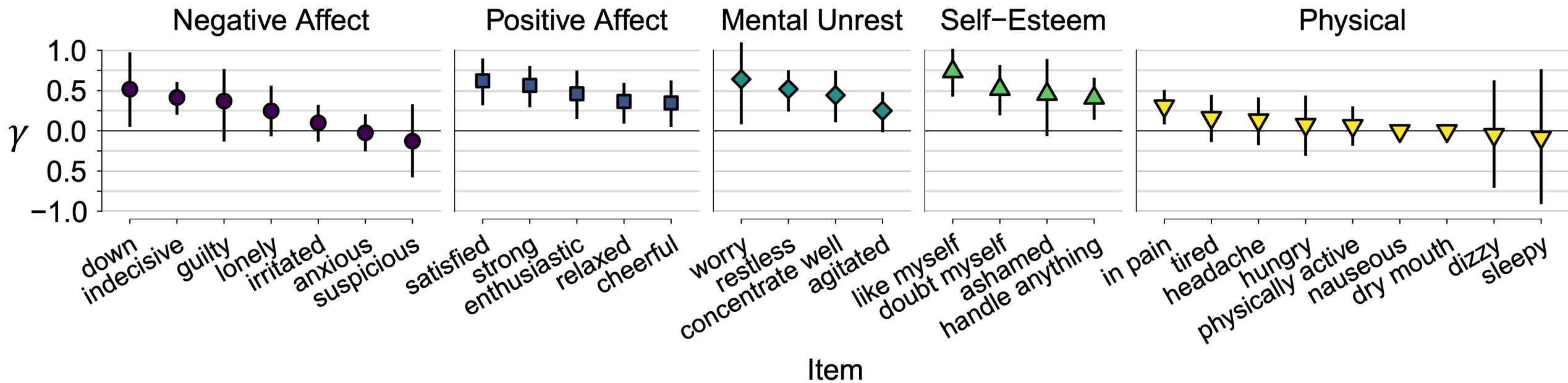
Model specification

$$\begin{aligned}y_{d,1} &= c + \gamma y_{d-1,B} + \varepsilon_{d,1} \\ y_{d,b} &= c + \phi y_{d,b-1} + \varepsilon_{d,b} \quad \text{for } b = 2, \dots, B\end{aligned}$$

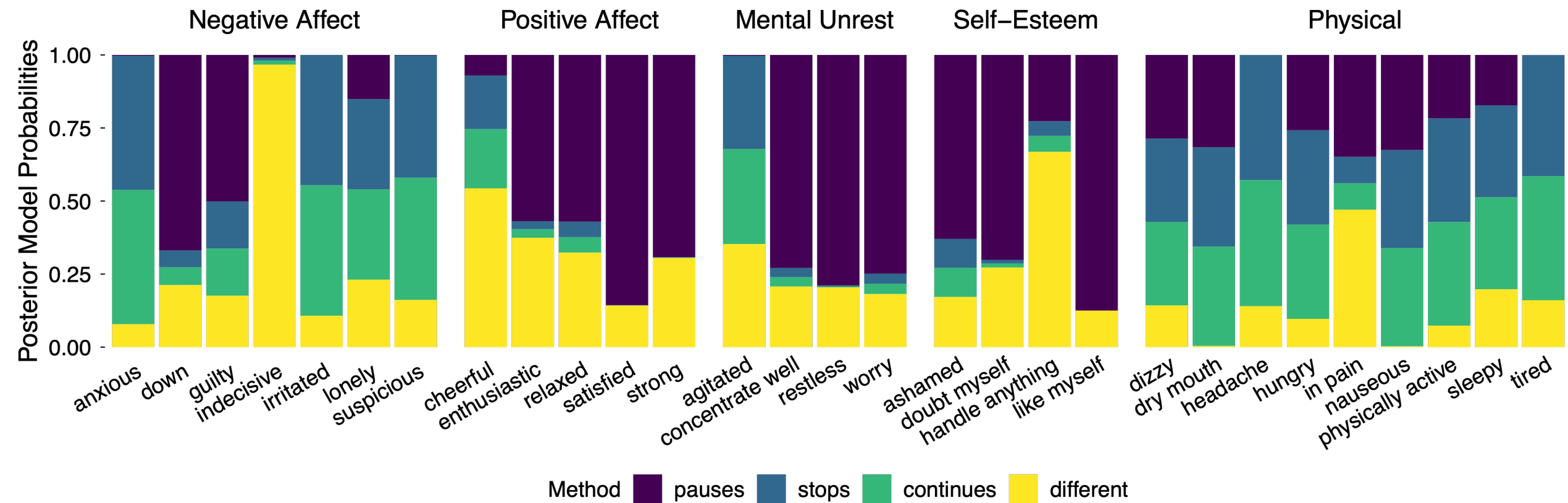
Or equivalently:

$$\begin{aligned}y_t &= c + \phi y_{t-1} + \chi y_{t-1} D_t + \varepsilon_t \\ \gamma &= \phi + \chi \\ D_t &= \begin{cases} 0 & \text{for } b = 2, \dots, B \\ 1 & \text{for } b = 1 \end{cases}\end{aligned}$$

Estimated nighttime autoregressions from the Peter Groot data



Model comparisons



Interval length correction sensitivity check

